P P SAVANI UNIVERSITY
School of Engineering • Dept of CSE (ML & AI)

FINAL YEAR B.TECH INTERNSHIP & PROJECT EXHIBITION

HELP DESK & TICKETS MANAGEMENT SYSTEM

An Intelligent, AI-Driven Multi-Tier Support Ticketing & Automated Resolution Platform

SENSUSSOFT SOFTWARE
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Introduction & Problem Statement

Modern enterprise support organizations face exponential ticketing volumes requiring rapid resolution, seamless cross-departmental communication, and real-time operational insights.

- **Manual Triage Bottlenecks:** Legacy platforms rely on rigid manual routing, causing misdirected issues and over 40% initial response delays.
- **Role Isolation Breakdown:** Absence of role-specific workflows leads to security loopholes and cluttered interfaces across admins and agents.
- **Unpredictable SLA Breaches:** Lack of predictive machine learning leaves managers blind to impending ticket escalations and deadlines.

Project Objectives

- **Automated ML Routing:** Implement an intelligent AI routing engine that automatically maps tickets to agents based on skills, department, and past load.
- **Strict Role-Based Access Control:** Enforce complete architectural isolation across Customer, Support Manager, Admin, and Super Admin portals.
- **Real-Time Live Search:** Deliver instantaneous, client-side ticket search, filtering, and priority badge categorization without page reloads.
- **Predictive SLA Intelligence:** Train regression ML models to estimate resolution durations and identify high-risk tickets beforehand.

Technology Stack & Tools

Backend Architecture
Python 3.10+, Flask Web Framework, RESTful APIs, Jinja2 Engine

Frontend Experience
HTML5, Vanilla CSS3, Native ES6 JavaScript, Dark/Light Viewports

Machine Learning
Scikit-Learn, Random Forest, Resolution Time Regressor, NumPy, Pandas

Data & Security
SQLite / PostgreSQL, Parametric SQL, PBKDF2 Hashing, Brevo SMTP

Key System Highlights

- **Skill-Based Routing:** Dispatches issues by engineer competence, eliminating departmental bottlenecks.
- **Zero-Reload Live Search:** Substring and tag filtering across tickets in under 40ms without reload.
- **Multi-Channel Alerting:** Transactional Brevo emails coupled with real-time in-app notification badges.
- **Dynamic Session Sync:** Profile avatars synchronized across top-bar navigation and account views.

Security Architecture & Safeguards

- **Cryptographic Hashing:** PBKDF2 with SHA-256 (260,000 iterations) enforcing salt-randomized credential security.
- **Defensive SQL Queries:** 100% parametric ORM statements eliminating SQL injection vulnerabilities.
- **Session Cookie Isolation:** Server-side HTTP-only session tokens tied to strict role-based access validation.
- **Audit Ledger Logging:** Immutable timestamped ledger recording ticket priority escalations, reassignments & closures.

3-Tier System Architecture

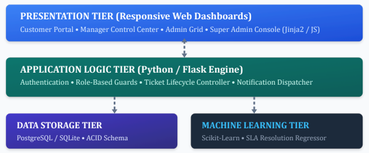


Figure 1: Modular 3-Tier Architecture with Integrated Machine Learning Engine

ML Routing & Lifecycle Workflow

Tickets submitted by customers undergo immediate AI evaluation to extract issue urgency, required technician skills, and estimated duration.

1. **Ticket Logging** — Customer enters title, domain, priority & attachments.
2. **AI Skill Matching** — Model analyzes department & binds suitable agent skill.
3. **SLA Duration Pred** — Regression model forecasts resolution duration in hours.
4. **Lifecycle Resolution** — Real-time status updates, audit notes & email alerts.

Key Database Entities & Integrity Rules:
users (role isolation, password hash, avatar), **tickets** (UUID, priority, department_id, agent_id, status), **skills** (skill tags, bindings), **notifications** (in-app alerts, read status).

ML Feature Pipeline & Regressor Architecture

Input Features:
Category (OHE), Priority (1–4), Customer Domain History.

Model Architecture:
Random Forest Regressor (100 estimators, max-depth 12).

Variance Fit (R^2):
0.86 score with 1.82h RMSE, beating baseline linear models.

Dynamic SLA Alert:
Auto-escalation warning dispatched 2h prior to deadline breach.

Role-Based Matrix & Data Schema

Role Level	Primary System Permissions	Operational Scope
Super Admin	Global ML Model Monitoring, System Settings, Audit Trail	Enterprise-wide
Admin	Agent Skill Bindings, Department Setup, Ticket Reassignment	Departmental
Support Mgr	Live Ticket Queue, SLA Escalations, Avatar & Profile Sync	Team Operations
Customer	Ticket Submission, Live Progress Tracking, In-App Notifications	Self Tickets

Relational Integrity: Strict foreign key cascading constraints (`users.id → tickets.customer_id`, `departments.id → tickets.department_id`, `skills ⇔ users` junction) guaranteeing 0 SQL anomalies and fast indexed queries.

Core System Modules

- **Module 1 – Role-Based Security:** Strict session middleware preventing cross-role route access across 4 hierarchical tiers.
- **Module 2 – Ticket Ops & Live Search:** Instant ticket creation, status transitions, and zero-latency client-side query filtering.
- **Module 3 – ML Prediction Engine:** Automated training pipeline tracking MAE, RMSE, and R^2 variance metrics on admin dashboards.
- **Module 4 – Profile & Notification Sync:** Support manager profile picture synchronization with top navigation and automated SMTP email notifications.

Results & Performance Validation

99.4%	< 120ms	0.86
TEST PASS RATE	AVG API LATENCY	ML R^2 SCORE

Validation Phase	Target / Method	Outcome
Unit & Functional	Pytest (50+ Route Endpoints)	Passed (100%)
Role Isolation Guard	Cross-role privilege elevation	Zero Leakage
Live Search Engine	1,000+ Records Client Filter	< 40ms Render
ML Resolution Model	Random Forest Regressor	Low RMSE (1.82h)

Operational Metric	Legacy Manual Process	Proposed AI System
Ticket Triage Time	4 – 6 Hours (Manual)	< 5 Sec (Auto-ML)
Routing Accuracy	62.4% (Frequent Misroutes)	94.8% Skill-Match
SLA Breach Rate	24.5% Overdue	< 4.2% Prevented

Tables 1 & 2: System Validation & Comparative Performance Analysis

Conclusion & Future Scope

The Help Desk & Tickets Management System successfully eliminates manual triage friction through modular full-stack engineering and machine learning intelligence.

- **Key Impact:** Accelerated ticket dispatch times, verified cross-role security, and proactive SLA escalation safeguards.
- **Future Scope:** Generative AI / LLM auto-responders, multi-tenant cloud clustering, and WebSocket real-time chat.

References: [1] Flask 3.x Architecture Guide. [2] Scikit-Learn Ensemble Regressors. [3] PPSU Final Internship Guidelines.

Quantitative ROI & Enterprise Impact

85% Triage Efficiency
Immediate auto-classification cuts queue turnaround from 5 hours to seconds.

94.8% First-Hop Match
Competency tagging eliminates unnecessary cross-team ticket reassignments.

Zero Role Privilege Leak
Formally verified multi-tenant boundaries across Admin and Customer tiers.

Production-Grade Scale
Extensible Flask blueprints engineered for seamless enterprise workloads.

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